Decision Trees

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Decision Trees before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Baselines First

Without opening the old lesson, explain one key idea from Baselines First. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Decision Trees trace and record the first step you can explain without help.

Use W2: Verify the candidate scores for **Fitting** a decision tree. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Decision Trees explanation against the supplied facts before discussing it.

Use the worked case for **Fitting** a decision tree. Proposed explanation: “A near **branch** must predict blocked”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A stored fitted rule says spots >=5 predicts spotted; otherwise healthy. Trace4,5,6 and a missing count. Policy: missing count requires review. Can test labels be used to choose this cutoff?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Decision Trees target to a trace and a justified conclusion.

Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Decision Trees case and show the deciding evidence.

A supplied tree predicts spotted for spot_count>=3, otherwise healthy.

Trace 2,3,4 and state boundary.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

> A tree predicts fern when **leaf**_width_cm <= 3 and grass otherwise. Trace widths 2, 3 and 4. Which case checks the boundary?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 11 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Decision Trees adds to Responsible Plant Classifier and keep the evidence.

Draw the current Plant Classifier tree and annotate feature, **threshold**, **prediction**, and test **evidence** at each **leaf**.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

> Rule: if spots>=4 predict spotted, otherwise healthy. Trace 3,4,5 and missing. Does applying this rule demonstrate that **training** found the **threshold**?

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Name the parts and decisions

> Name the parts and decisions. Apply the supplied Decision Trees facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Decision Trees", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Identify the **root**, its two **branch**es, **leaf prediction**s and **depth** of the worked fitted tree. Which parts were selected using **training** labels?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Verify the candidate scores

Verify the candidate scores. Apply the supplied Decision Trees facts and keep your reasoning.

For each near **branch** and each tall **branch**, count blocked and clear labels and choose a **leaf**. Give the total errors for each candidate question.

My reasoning and evidence

W3 Separate fitting from prediction

> Separate fitting from prediction. Apply the supplied Decision Trees facts and keep your reasoning.

> **Where to find the supplied example**
>
> Reference for this exercise: open this week's student lesson, "Decision Trees", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

A learner sees a blocked validation case with near=0 and changes that **leaf** to blocked before reporting the score. Explain what changed and how to report the original fitted tree fairly.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Fit a new tree independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Fit a new tree independently. Apply the supplied Decision Trees facts and keep your reasoning.

Rules supplied for this independent case

Rules for this fresh tree: try near (feature 0) before tall (feature 1). Each depth-one candidate splits on 0 versus 1 and predicts the training-majority label within each leaf; a tied leaf predicts clear. Select the fewest total training mistakes, retaining the first candidate on a tie. Fit the baseline to the whole training-majority label, with clear on a tie. Validation never changes either fitted rule.

New **training** rows (near,tall,label) are (1,0,clear),(1,1,clear),(1,0,clear),(0,1,blocked),(0,0,blocked),(0,1,clear). Use the same candidate order and tie policies. Give both candidates' leaves and errors, then select the tree.

Validation rows are (1,1,clear),(0,0,blocked),(0,1,clear),(1,0,blocked). List tree **prediction**s and compare accuracy with the **training**-majority baseline.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Audit a tie and a claim

> Audit a tie and a claim. Apply the supplied Decision Trees facts and keep your reasoning.

If near and tall make the same **training** error count, which question does this code retain and why? Does retaining that question prove it will be better on future trips?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Decision Trees

> Trace a small fitted tree, including its exact boundary. Explain the labelled step before

> checking the changed condition.

Trace a small fitted tree, including its exact boundary.

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/11

Decision Trees

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read first branch

L: 1>=2 is false, so predict healthy.



2. Read boundary

M: 2>=2 is true, so predict spotted.



3. Stop missing path

N lacks a count, so use review rather than inventing zero.

CHANGE ONE THING

The stored rule changes to visible_spots>2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Trace a small fitted tree, including its exact boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? N lacks a count, so use review rather than inventing zero.

1. Read boundary

2. Read first **branch**

3. Stop missing path

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop missing path N lacks a count, so use review rather than inventing zero. Read the picture in order. N lacks a count, so use review rather than inventing zero.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The stored rule changes to visible_spots>2. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. M is now predicted healthy.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M is now predicted healthy. Changing >= to > moves the exact value 2 to the other **branch**.

Repair a missing building block

Review Baselines First (ai-4/10). Goal: Fit a majority label baseline using **training** labels only. Short practice: Can the candidate's **training** accuracy replace this validation comparison?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Leakage (ai-4/8). Goal: Choose a **split** unit that matches the intended **prediction** claim. Short practice: How many independently new plant identities are in the original test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Decision Trees picture from its labels, then record a relevant limit.

01. Decision Trees: the case

AI ACADEMY / AI-4/11 · 01 · LOOK AT THE FACTS

Decision Trees: the case

The situation

Trace a small fitted tree, including its exact boundary.

Supplied evidence

Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. Inputs L=1, M=2, N=None.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain decision trees using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

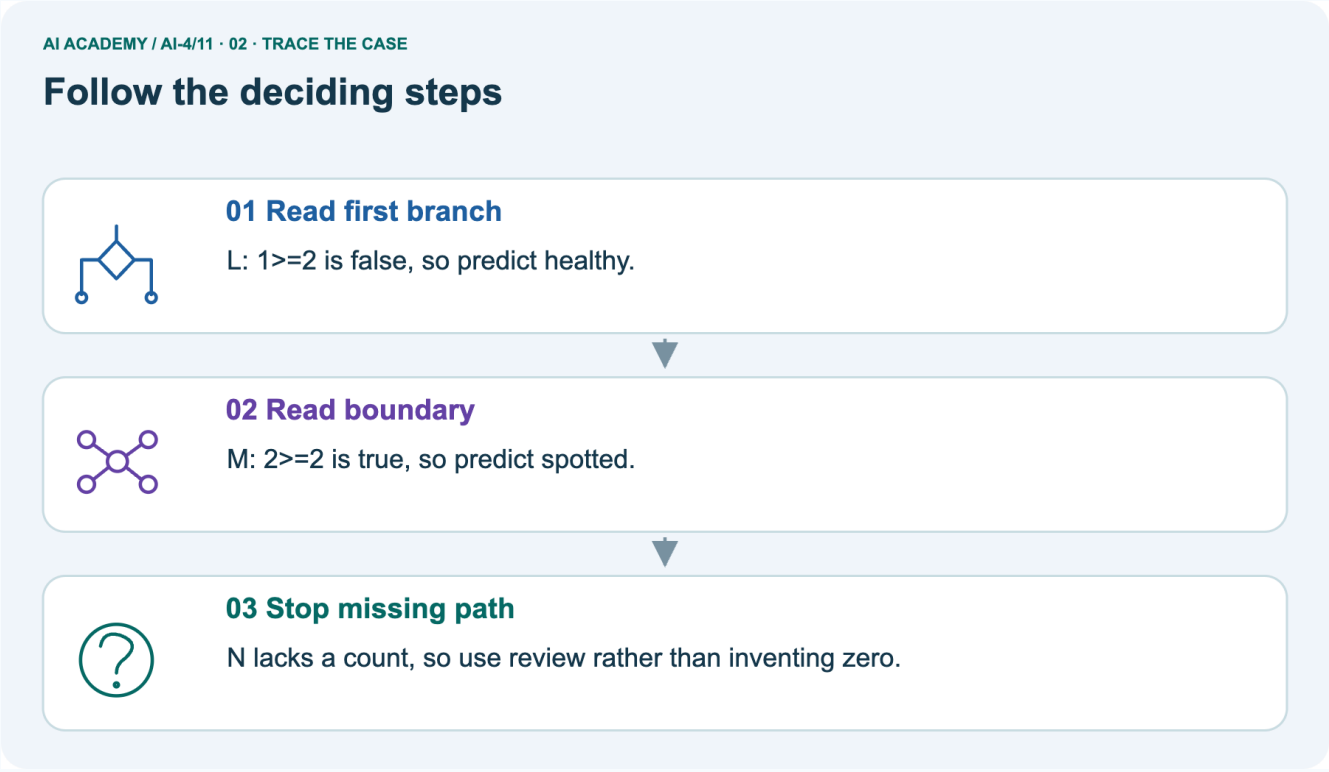
Trace a small fitted tree, including its exact boundary. Supplied facts: Toy stored tree: if visible_spots>=2 predict spotted; otherwise healthy. Missing visible_spots requires review. **Input**s L=1, M=2, N=None.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



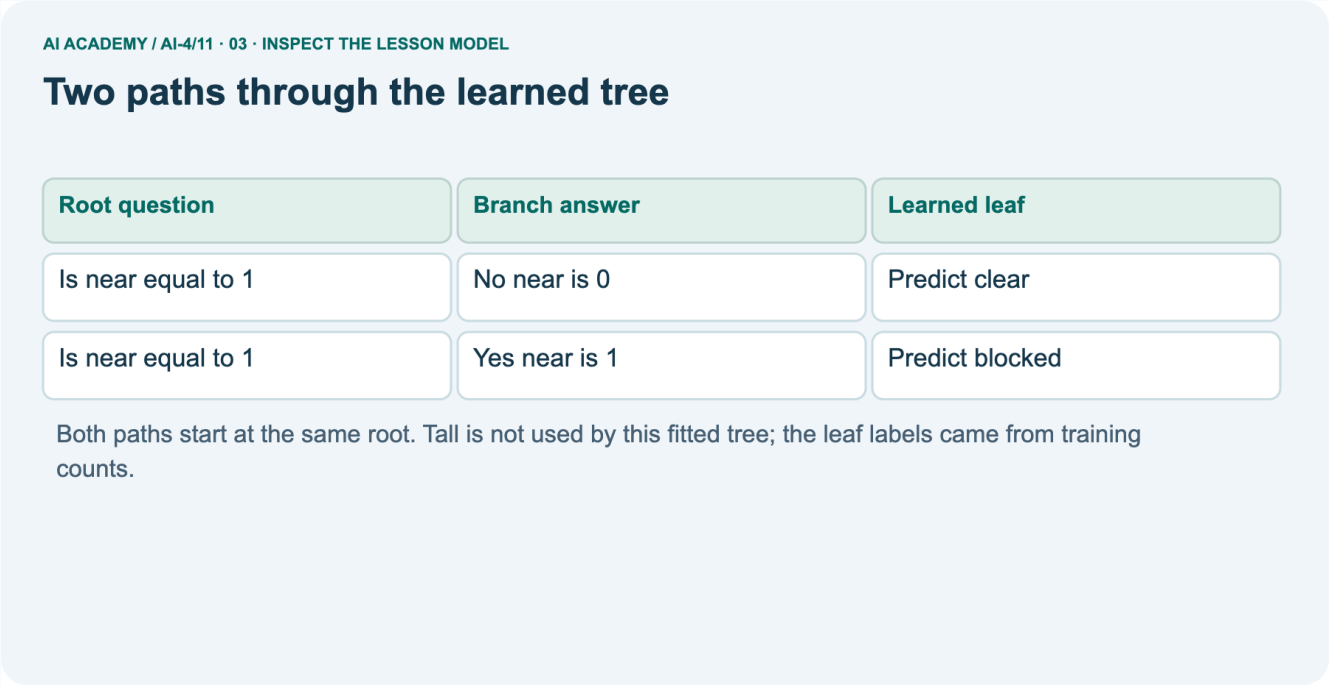
Follow the labelled picture with the matching explanation. Open the vector diagram

Read first **branch**: L: 1>=2 is false, so predict healthy. → Read boundary: M: 2>=2 is true, so predict spotted. → Stop missing path: N lacks a count, so use review rather than inventing zero.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two paths through the learned tree



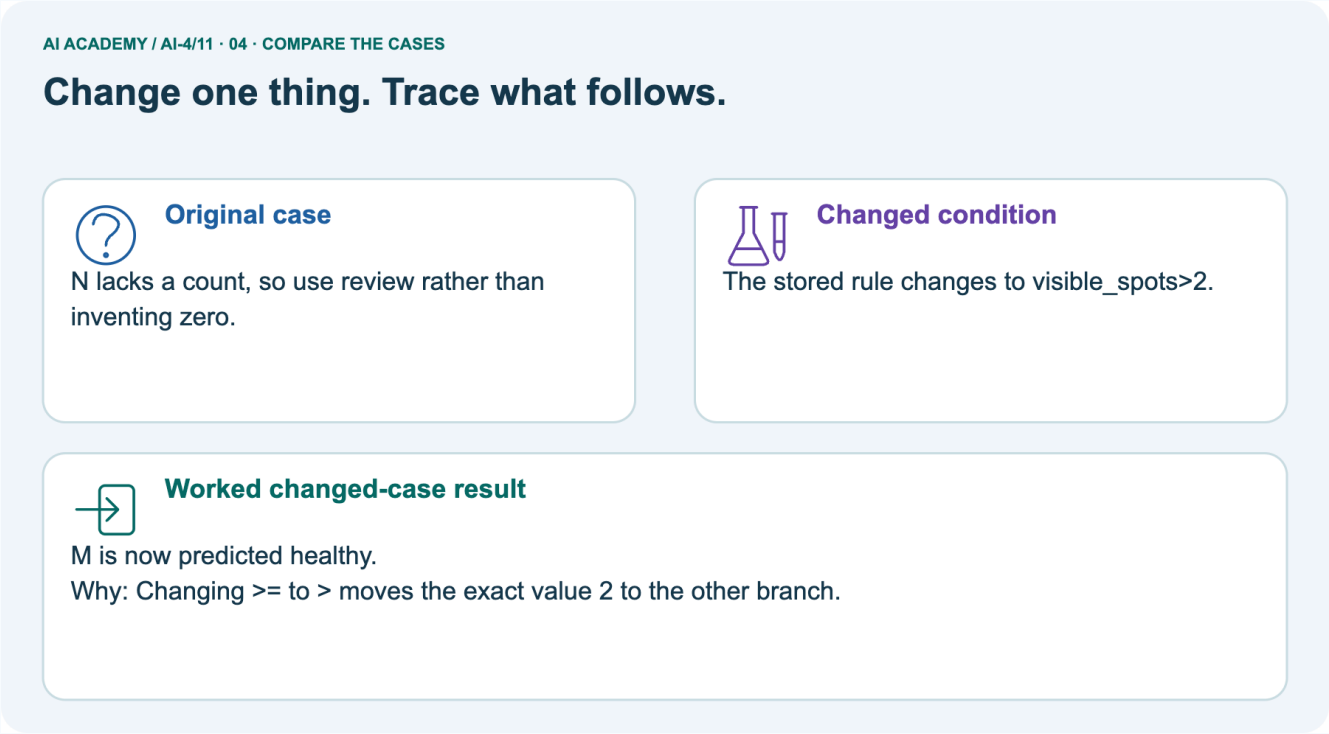
Follow the labelled picture with the matching explanation. Open the vector diagram

Two paths through the learned tree. Is near equal to 1 | No near is 0 | Predict clear. Is near equal to 1 | Yes near is 1 | Predict blocked

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

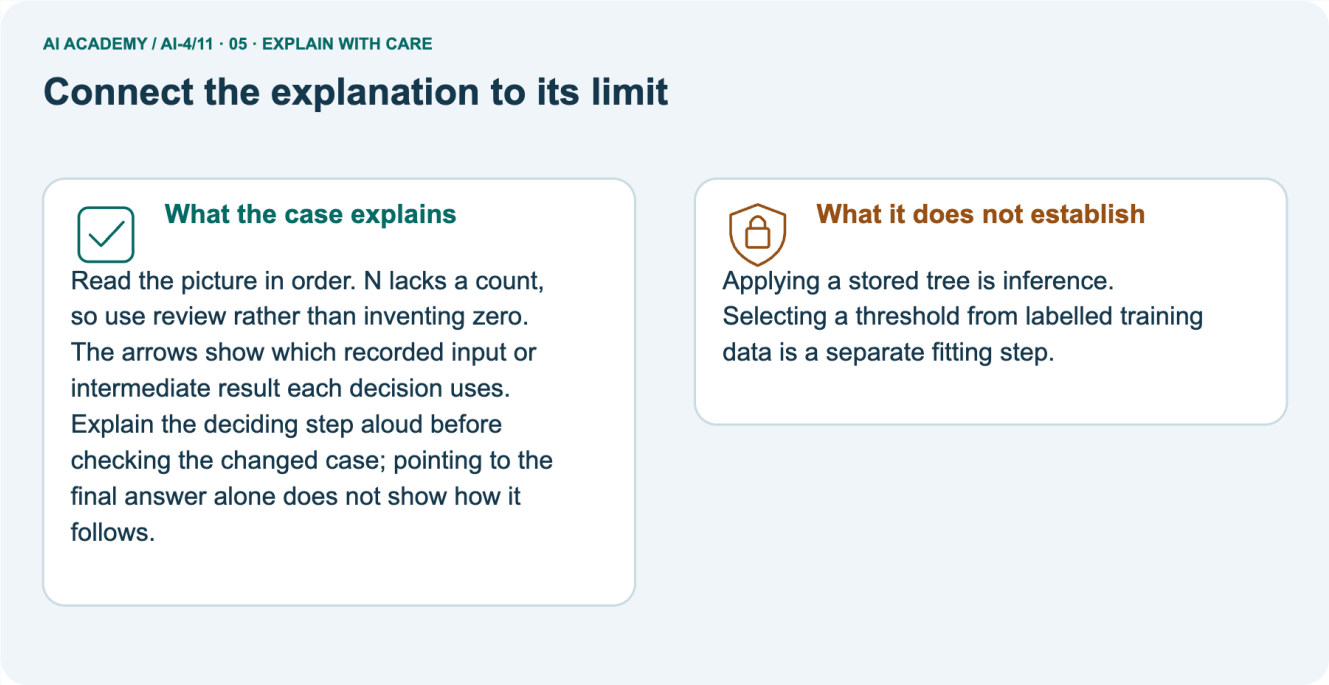
Original case: N lacks a count, so use review rather than inventing zero. Changed condition: The stored rule changes to visible_spots>2. Expected result: M is now predicted healthy.

Reason: Changing >= to > moves the exact value 2 to the other **branch**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. N lacks a count, so use review rather than inventing zero. The arrows show which recorded **input** or intermediate result each decision uses.

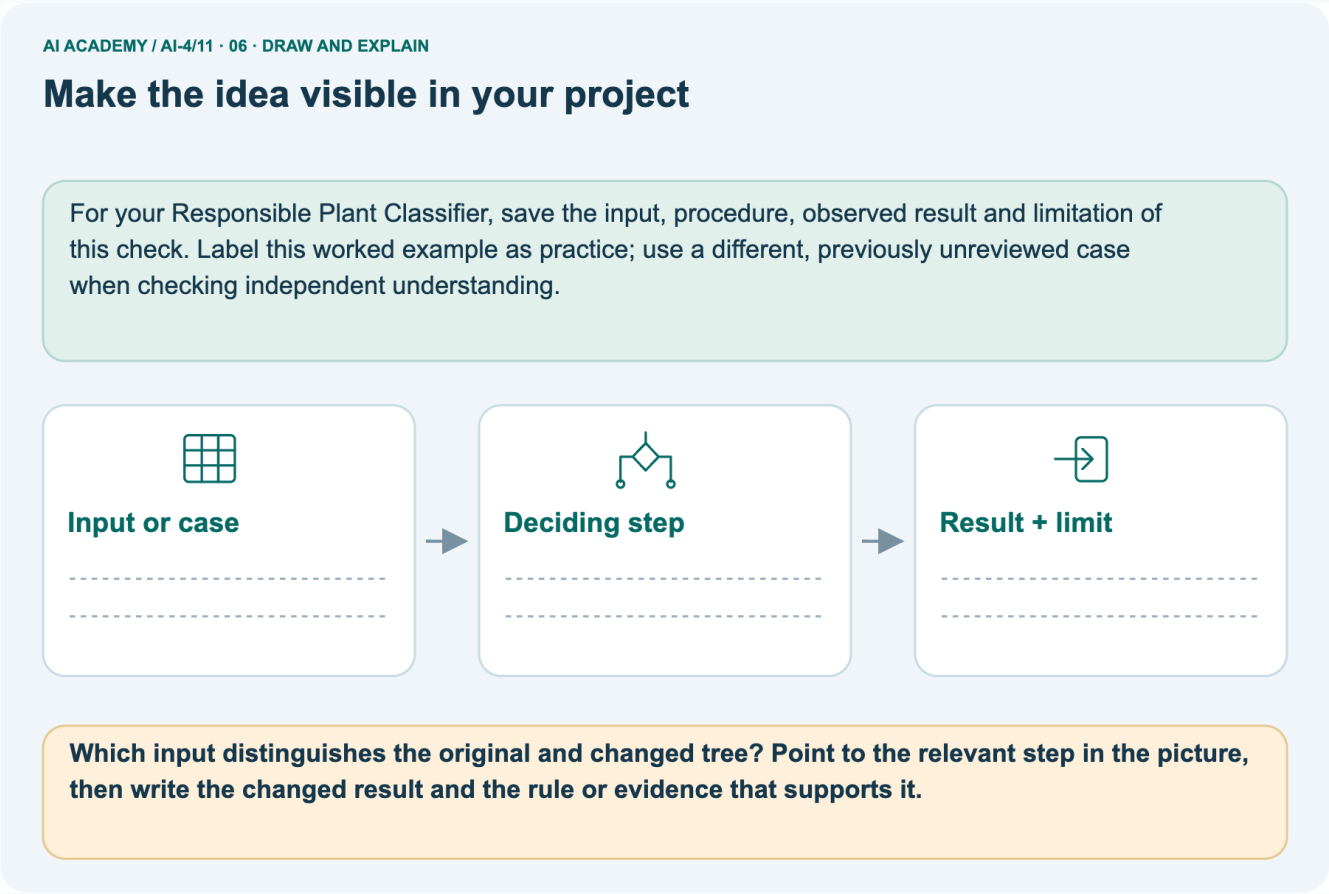
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Applying a stored tree is inference.

Selecting a **threshold** from labelled **training** data is a separate **fitting** step.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** distinguishes the original and changed tree? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.